

States Of Matter

1. A gas mixture containing 160 parts per million by volume of H_2S in Ne has a total pressure of 725 mmHg. What is the partial pressure of H_2S ?

(A) 0.069 mmHg (B) 0.12 mmHg
(C) 0.16 mmHg (D) 0.20 mmHg

2. Which gas has the highest root-mean-square velocity at 298 K?

(A) CO (B) CO_2 (C) CH_4 (D) CF_4

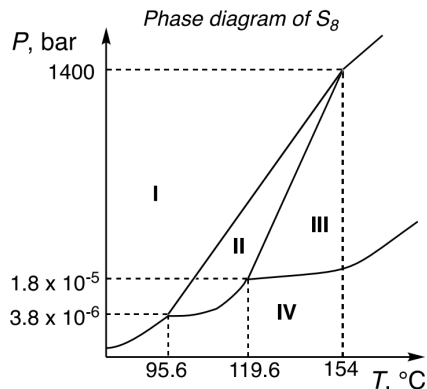
3. Which statement describes a change when a pure solid melts spontaneously?

(A) The entropy decreases.
(B) The free energy increases.
(C) The vapor pressure increases.
(D) The process is endothermic.

4. Which compound has the highest vapor pressure at 25 °C?

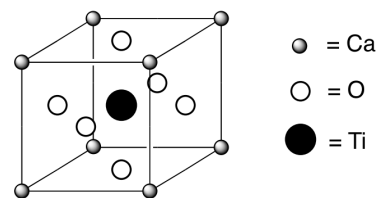
(A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (B) $\text{CH}_3\text{CH}_2\text{OCH}_3$
(C) $\text{CH}_3\text{CHOHCH}_3$ (D) $\text{HOCH}_2\text{CH}_2\text{OH}$

5. Sulfur has two stable solid phases, monoclinic and orthorhombic sulfur. Which regions of the phase diagram correspond to these two solid phases?



(A) I and II (B) I and IV
(C) II and III (D) II and IV

6. The mineral perovskite has the cubic unit cell shown below, with the oxygens occupying the centers of the faces, the titanium in the center, and the calciums on the corners of the cube. What is the formula of perovskite?



(A) CaTiO_3 (B) Ca_2TiO_4
(C) CaTi_2O_4 (D) Ca_8TiO_6

7. 1.53 g of a gas occupies a volume of 264 mL at 722 mm Hg and 22 °C. What is the molecular formula of the gas?

(A) C_2F_6 (B) C_3HF_5
(C) $\text{C}_3\text{HF}_5\text{O}$ (D) $\text{C}_3\text{F}_6\text{O}$

8. A mole of which gas will have a volume closest to 22.4 L at 0 °C and 1 atm pressure?

(A) H_2 (B) CO_2
(C) SF_6 (D) Xe

9. Two identical steel cylinders at the same temperature contain equal masses of CO(g) and $\text{C}_2\text{H}_2\text{(g)}$. Which quantities will be greater for the cylinder containing C_2H_2 ?

I. Pressure II. Average molecular velocity

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

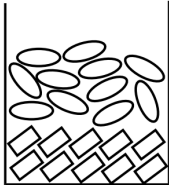
10. The critical point of carbon dioxide is 31.0 °C and 72.8 atm. Which best describes the composition of a sample of pure CO_2 at 29.0 °C and 75 atm?

(A) Only gaseous CO_2 is present.
(B) Only liquid CO_2 is present.
(C) A mixture of gaseous CO_2 and liquid CO_2 is present.
(D) Only supercritical CO_2 is present.

11. The normal boiling point of ethane (C_2H_6) is -88.5 °C, while the normal boiling point of fluoromethane (CH_3F) is -78.4 °C. A difference in which type of intermolecular forces is most responsible for the higher boiling point of fluoromethane?

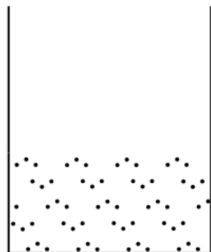
(A) London dispersion forces
(B) Dipole-dipole interactions
(C) Ion-dipole interactions
(D) Hydrogen bonding

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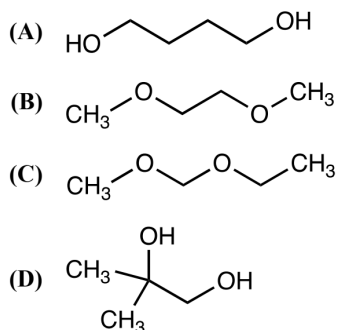
12. Cadmium metal crystallizes in a hexagonal closest packed structure. How many nearest neighbors does a cadmium atom have in this structure?
- (A) 4 (B) 6 (C) 8 (D) 12
13. A white crystalline solid melts at 400 °C and dissolves in water to give an electrically conducting solution. Which is it?
- (A) LiOH (B) C₁₂H₂₂O₁₁
(C) SiO₂ (D) Cu
14. A 10.0 L vessel containing 5.0 atm of He and a 3.0 L vessel containing 10.0 atm of Ar are connected by a valve. The valve is opened and the gases are allowed to mix at constant temperature. What is the final pressure in the vessels?
- (A) 2.3 atm (B) 3.8 atm
(C) 6.2 atm (D) 7.5 atm
15. A mixture of **A** (whose molecules are represented by rectangles) and **B** (whose molecules are represented by ovals) is represented schematically below. Which is the best description of this mixture?
- 
- (A) **A** is dissolved in liquid **B**.
(B) **A** is a crystalline solid in the presence of liquid **B**.
(C) **A** and **B** are crystalline solids with **A** denser than **B**.
(D) **A** and **B** are immiscible liquids with **A** denser than **B**.
16. Which statement best describes a sample of ammonia at equilibrium at its triple point (195.4 K, 0.0606 bar)?
- (A) Gaseous ammonia must be present, and solid or liquid ammonia may be present.
(B) Liquid ammonia must be present, and solid or gaseous ammonia may be present.
(C) Solid ammonia must be present, and liquid or gaseous ammonia may be present.
(D) Gaseous, liquid, and solid ammonia must all be present.
17. Which compound's lattice energy is largest in magnitude?
- (A) LiF (B) NaCl (C) MgO (D) CaS
18. Tantalum crystallizes in a body-centered cubic structure. How many nearest neighbors does each tantalum atom have?
- (A) 4 (B) 6 (C) 8 (D) 12
19. A room temperature sample of xenon gas in a metal container with a piston has a volume of 1.00 L and a pressure of 4.0 atm. The piston is moved to change the volume of the container to 2.00 L, after which the pressure reads 1.75 atm. Which is the best explanation for this observation?
- (A) The volume is miscalibrated and the actual final volume is 1.75 L.
(B) Some xenon has leaked out through the seal in the piston.
(C) The friction of moving the piston has caused the temperature of the gas to increase.
(D) The system is correctly calibrated, well sealed, and remains at room temperature.
20. In which are the root mean square velocities of Xe(g), Ne(g), and N₂(g) at 290 K and 3.0 atm ranked from lowest to highest?
- (A) Xe < Ne < N₂ (B) Xe < N₂ < Ne
(C) N₂ < Ne < Xe (D) Ne < N₂ < Xe
21. In a sample of pure silane (SiH₄), which intermolecular forces are present?
- I. London dispersion forces
II. Hydrogen bonding
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
22. The melting point of an organic compound X is 5.0 °C. How do the vapor pressures of X(s) and X(l) compare at 10.0 °C?
- (A) The vapor pressure of X(s) is less than the vapor pressure of X(l).
(B) The vapor pressure of X(s) is equal to the vapor pressure of X(l).
(C) The vapor pressure of X(s) is greater than the vapor pressure of X(l).
(D) The relative vapor pressures cannot be determined without knowledge of the relative densities of X(s) and X(l).

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23. The arrangement of atoms (represented as black dots) in a pure substance contained in a beaker is shown schematically below. Which is the best description of the phase of the substance?



- (A) Crystalline molecular solid
(B) Crystalline network covalent solid
(C) Amorphous network covalent solid
(D) Liquid
24. Which statement about a sample of ammonia (NH_3) at its triple point is correct?
- (A) It has a pressure of 1.0 atm.
(B) It can be at only one possible temperature.
(C) There are equal amounts of NH_3 in the solid, liquid, and gas phases.
(D) If it is in a closed, well-insulated container, it will remain at the triple point when the volume of the container is decreased by a factor of two.
25. What is the partial pressure of Ar in a gaseous mixture of 3.25 g Ar and 2.28 g Kr in a 5.00 L container at 15.0 °C?
- (A) 0.0572 bar (B) 0.390 bar
(C) 0.412 bar (D) 0.447 bar
26. Which isomer of $\text{C}_4\text{H}_{10}\text{O}_2$ has the highest normal boiling point?

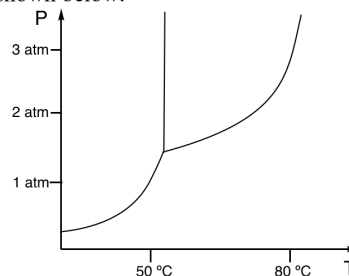


27. Which statements about the behavior of gaseous H_2 molecules in a container at 1 atm and 298 K are correct?

- I. All H_2 molecules are moving at the same speed.
II. The H_2 molecules are colliding more frequently with the walls of the container than they would in the same container at 398 K.

- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

28. A portion of the phase diagram of uranium hexafluoride (UF_6) is shown below.

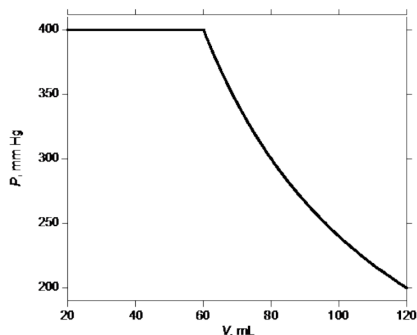


Which statements are correct?

- I. UF_6 sublimes, rather than melts, at 1 atm pressure.
II. At 2 atm and 45 °C, only solid UF_6 is present at equilibrium.
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
29. How many nearest neighbors does each silicon atom have in a crystal of Si?
- (A) 4 (B) 6
(C) 8 (D) 12

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30. A sample of methanol, CH_3OH , is introduced into an evacuated chamber with a movable piston. The pressure is measured as a function of the volume of the container while the temperature is maintained at 50°C , and the graph below is obtained. Which statement best describes this system?

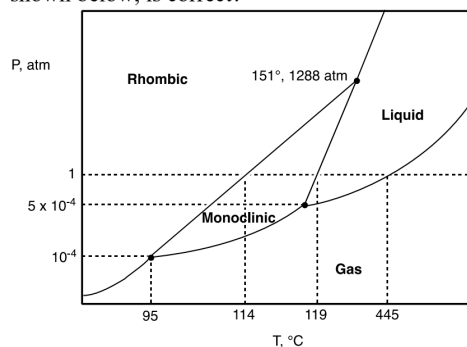


- (A) At volumes below 60 mL, methanol is only present in the liquid phase.
 (B) At volumes above 60 mL, methanol is only present in the gaseous phase.
 (C) If the process were repeated at 60°C , the point at which the pressure begins to decrease would shift to higher volume.
 (D) If more methanol were added to the container, the pressure at $V = 20$ mL would increase.
31. When liquid water converts to solid ice at 0°C and 1 atm pressure, which of the following changes takes place?
- (A) Heat flows out of the system.
 (B) The volume of the system decreases.
 (C) The vapor pressure of water increases.
 (D) The hydrogen bonds become weaker.
32. Which isomer of $\text{C}_4\text{H}_8\text{O}_2$ has the greatest solubility in water?
- (A)
- (B)
- (C)
- (D)
33. Which compound has the lowest normal boiling point?
- (A) HF (B) HCl (C) HBr (D) HI

34. A sealed container contains a gaseous sample at 300.0 K consisting of either pure ethane (C_2H_6), pure neon (Ne), or a mixture of the two. The pressure inside the container at this temperature is 1.00 atm . When the container is cooled to 150.0 K , the pressure is 0.37 atm . What is the composition of the sample? (The vapor pressure of C_2H_6 at 150 K is 0.10 atm ; the critical temperature of Ne is 44 K .)

- (A) Pure ethane
 (B) Pure neon
 (C) A mixture of ethane and neon.
 (D) It cannot be determined from the information given.

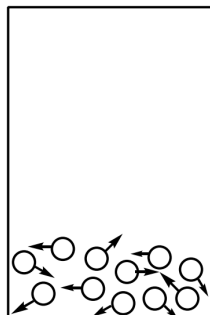
35. Which statement about sulfur, whose phase diagram is shown below, is correct?



- (A) Monoclinic sulfur sublimates at 1 atm pressure while rhombic sulfur does not.
 (B) Rhombic sulfur is less dense than monoclinic sulfur.
 (C) At temperatures above 119°C , monoclinic sulfur cannot exist at equilibrium.
 (D) The conversion of rhombic sulfur to monoclinic sulfur is endothermic.
36. Aluminum crystallizes in a face-centered cubic unit cell with an edge length of 405 pm . What is its density?
- (A) 0.674 g cm^{-3} (B) 1.35 g cm^{-3}
 (C) 2.70 g cm^{-3} (D) 5.40 g cm^{-3}

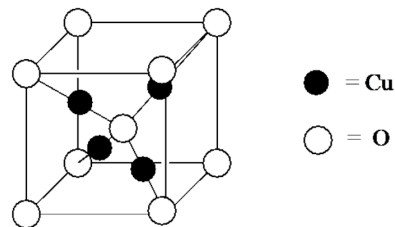
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37. The diagram is a microscopic view of a snapshot of a substance at equilibrium, with the circles representing molecules and the arrows the molecules' velocities. What state of matter is depicted?

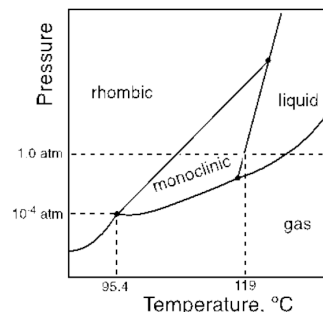


- (A) Solid (B) Liquid (C) Gas (D) Plasma
38. The vapor density of which fluorocarbon is 6.17 g L^{-1} at 23°C and 1.00 atm ?
 (A) C_3F_6 (B) C_3F_8 (C) C_4F_6 (D) C_4F_8
39. What is the principal intermolecular force that must be overcome when *n*-hexane (C_6H_{14} , bp = 69°C) is vaporized?
 (A) Hydrogen bonding
 (B) Covalent bonding between carbon atoms
 (C) Dipole-dipole forces
 (D) London dispersion forces
40. The melting point of water decreases with increasing pressure. Which is the best explanation for this observation?
 (A) Liquid water is denser than solid water at 0°C .
 (B) Melting of ice is endothermic at 0°C .
 (C) The vapor pressure of liquid water is lower than the vapor pressure of solid water at 0°C .
 (D) Solid and liquid water cannot coexist at equilibrium at 0°C at pressures different from 1 atm .
41. A cylinder containing a mixture of CO and CO_2 has a pressure of 2.00 atm at 93°C (366 K). The cylinder is then cooled to -90°C (183 K), where CO is still a gas but CO_2 is a solid with a vapor pressure of 0.25 atm . The pressure in the cylinder at this temperature is 0.90 atm . What is the mole fraction of CO_2 in the cylinder?
 (A) 0.10
 (B) 0.28
 (C) 0.35
 (D) It cannot be determined from the information given.

42. A mineral containing only copper and oxygen adopts the cubic unit cell shown below. What is the formula of this mineral?



- (A) Cu_2O (B) CuO (C) Cu_3O_2 (D) Cu_4O_9
43. Which gas has an effusion rate closest to half that of oxygen gas at a given temperature?
 (A) H_2 (B) He (C) SO_2 (D) Xe
44. Which compound has the highest normal boiling point?
 (A) 1-butanol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 (B) 2-butanol, $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{OH}$
 (C) 2-methyl-1-propanol, $(\text{CH}_3)_2\text{CHCH}_2\text{OH}$
 (D) 2-methyl-2-propanol, $(\text{CH}_3)_3\text{COH}$
45. A portion of the phase diagram of elemental sulfur is shown below. Which statement about sulfur is correct?



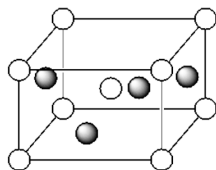
- (A) Rhombic sulfur cannot be sublimed without first converting to monoclinic sulfur.
 (B) The conversion of rhombic sulfur to monoclinic sulfur is exothermic.
 (C) Rhombic sulfur is denser than monoclinic sulfur.
 (D) At atmospheric pressure, monoclinic sulfur cannot be in equilibrium with liquid sulfur.
46. In which are the ionic compounds listed in order of increasing magnitude of their lattice energy?
 (A) $\text{CaO} < \text{RbI} < \text{LiF} < \text{LiI}$
 (B) $\text{RbI} < \text{CaO} < \text{LiI} < \text{LiF}$
 (C) $\text{LiI} < \text{LiF} < \text{CaO} < \text{RbI}$
 (D) $\text{RbI} < \text{LiI} < \text{LiF} < \text{CaO}$

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47. Sodium halides crystallize in the rock salt structure, which has a cubic unit cell with four formula units per unit cell. A sodium halide has a density of 2.83 g cm^{-3} and a unit cell edge length of 462 pm. Which compound is it?

(A) NaF (B) NaCl (C) NaBr (D) NaI

48. The tetragonal unit cell of white tin is shown below. Each circle represents a tin atom, with the gray circles sitting on the four vertical faces of the unit cell. How many tin atoms are contained in this unit cell?



(A) 2 (B) 4 (C) 8 (D) 13

49. Which gas at 600 K has the same effusion rate as methane (CH_4) at 150 K?

(A) He (B) O_2 (C) SO_2 (D) N_2O_4

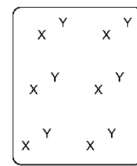
50. Ammonia (NH_3) has a higher normal boiling point (-33°C) than its heavier congeners PH_3 (bp -88°C) or AsH_3 (bp -63°C). Which is the best explanation for this difference?

- (A) NH_3 is trigonal pyramidal and polar while PH_3 and AsH_3 are trigonal planar and nonpolar.
 (B) NH_3 is much more acidic than PH_3 or AsH_3 .
 (C) $\text{NH}_3(l)$ experiences stronger London dispersion forces than $\text{PH}_3(l)$ or $\text{AsH}_3(l)$.
 (D) $\text{NH}_3(l)$ has extensive hydrogen bonding while $\text{PH}_3(l)$ and $\text{AsH}_3(l)$ do not.

51. In which are the ionic solids ranked in order of increasing melting point?

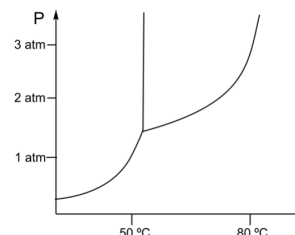
- (A) $\text{KBr} < \text{NaCl} < \text{NaF} < \text{MgO}$
 (B) $\text{NaF} < \text{NaCl} < \text{MgO} < \text{KBr}$
 (C) $\text{KBr} < \text{NaCl} < \text{MgO} < \text{NaF}$
 (D) $\text{MgO} < \text{NaF} < \text{KBr} < \text{NaCl}$

52. A two-dimensional slice through the lattice of a crystalline solid containing two different elements X and Y is shown schematically and to scale below. What type of solid is it?



- (A) A metallic alloy such as FeCr
 (B) A molecular solid such as IBr
 (C) An ionic compound such as LiCl
 (D) A network covalent solid such as SiC

53. A portion of the phase diagram of uranium hexafluoride (UF_6) is shown below.

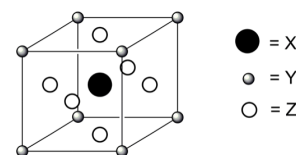


Which statements are correct?

- I. UF_6 sublimates at atmospheric pressure.
 II. At 80°C and 1.5 atm, only $\text{UF}_6(g)$ is present at equilibrium.

- (A) I only (B) II only
 (C) Both I and II (D) Neither I nor II

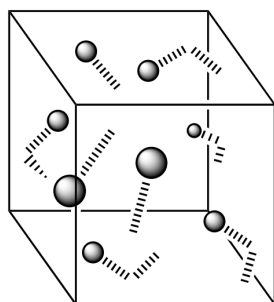
54. One unit cell of a crystal containing elements X, Y, and Z is shown below. What is its formula?



- (A) XYZ (B) XYZ_3 (C) XY_4Z_2 (D) XY_8Z_6

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55. What state of matter corresponds to the diagram below?



- (A) Gas (B) Liquid
(C) Amorphous solid (D) Crystalline solid
56. The normal boiling point of 2-propanol, $(\text{CH}_3)_2\text{CHOH}$, is 83°C , while that of acetone, $(\text{CH}_3)_2\text{C}=\text{O}$, is 56°C . What is the principal reason for the greater boiling point of 2-propanol?
- (A) The O–H bond in 2-propanol is stronger than the C–H bonds in acetone.
(B) 2-Propanol experiences greater London dispersion forces than acetone.
(C) 2-Propanol experiences stronger dipole-dipole interactions than acetone.
(D) 2-Propanol experiences stronger hydrogen bonding than acetone.
57. Consider the four gases CO_2 , N_2 , CCl_4 , and He. Which is the correct order of increasing average molecular speed at 100°C ?
- (A) $\text{He} < \text{N}_2 < \text{CO}_2 < \text{CCl}_4$
(B) $\text{CCl}_4 < \text{CO}_2 < \text{N}_2 < \text{He}$
(C) $\text{He} < \text{CO}_2 < \text{N}_2 < \text{CCl}_4$
(D) $\text{CCl}_4 < \text{N}_2 < \text{CO}_2 < \text{He}$
58. Under what conditions will the behavior of a real gas best approximate the behavior of an ideal gas?
- I. High temperature II. High pressure
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II
59. A solid has a melting point of 1710°C , is soluble in water, and does not conduct electricity in the solid state. What is the most likely nature of the bonding in this solid?
- (A) Molecular covalent (B) Network covalent
(C) Ionic (D) Metallic

60. Toluene, C_7H_8 , has both a higher vapor pressure than water at 25°C and a higher normal boiling point. Which statement best explains these observations?
- (A) Liquids with higher vapor pressures typically have higher boiling points.
(B) Toluene has a higher molar mass than water.
(C) Toluene has a lower heat of vaporization than water.
(D) The density of toluene vapor is greater than that of water vapor.
61. A sample of neon gas is held at 25.0°C and 1.0 atm in a cylinder with a movable piston. Under these conditions the gas occupies 5.0 L. What volume does the gas occupy at 12.5°C and 1.0 atm?
- (A) 2.5 L (B) 4.8 L (C) 5.2 L (D) 10 L
62. The normal boiling points of molecular fluorine, chlorine, bromine, and iodine increase in that order. Which of the following statements accounts for this increase?
- (A) The chemical reactivity decreases in that order.
(B) The London dispersion forces increase in that order.
(C) The dipole-dipole forces increase in that order.
(D) The hydrogen bonding increases in that order.
63. Quartz, SiO_2 , is the most common mineral found on the surface of the earth. What is the best explanation for the fact that quartz is hard and has a high melting point?
- (A) Quartz crystals are extended structures in which each atom forms strong covalent bonds with all of its neighboring atoms.
(B) Quartz crystals consist of positive and negative ions that are attracted to one another.
(C) Quartz crystals are formed under extremes of temperature and pressure.
(D) Silicon and oxygen atoms are especially hard because of their electronic structure.
64. The critical point of carbon dioxide is 304 K and 73 atm. Under which conditions is carbon dioxide a liquid?
- I. 303 K and 73 atm II. 305 K and 74 atm
- (A) I only (B) II only
(C) Both I and II (D) Neither I nor II

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65. The lattice energy (energy required to separate the ions in an ionic solid) of MgO is much larger than that of LiF. What contributes the most to this difference?

(A) Mg^{2+} is a smaller ion than Li^+ , and O^{2-} is a smaller ion than F^- .
(B) F is more electronegative than O, and Li is more electropositive than Mg.
(C) MgO contains doubly charged ions, while LiF contains singly charged ions.
(D) MgO contains more electrons than LiF.

66. Which best describes the bonding in Cu(s)?

(A) The copper atoms are positively charged in a sea of delocalized electrons.
(B) The copper atoms are alternately positively and negatively charged.
(C) The copper atoms form covalent bonds to adjacent copper atoms.
(D) The copper atoms form hydrogen bonds to adjacent copper atoms.

67. A sample of O_2 gas at a given temperature and pressure has a density of 1.30 g L^{-1} . What is the density of gaseous propane (C_3H_8) under the same conditions?

(A) 1.30 g L^{-1} (B) 1.79 g L^{-1}
(C) 1.96 g L^{-1} (D) 2.60 g L^{-1}

68. Which statements about the behavior of gaseous H_2 molecules in a container at 1 atm and 298 K are correct?

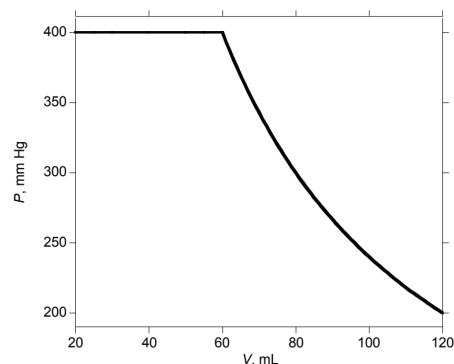
I. All H_2 molecules are moving at the same speed.
II. The H_2 molecules are colliding more frequently with the walls of the container than they would in the same container at 398 K.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

69. Which species has a normal boiling point closest to the normal boiling point of argon, Ar?

(A) H_2 (B) N_2 (C) F_2 (D) Cl_2

70. A sample of methanol, CH_3OH , is introduced into an evacuated chamber with a movable piston. The pressure is measured as a function of the volume of the container while the temperature is maintained at 50°C , and the graph below is obtained.



Which statements are correct?

- I. At volumes less than 60 mL, only liquid methanol is present.
II. At volumes greater than 60 mL, only gaseous methanol is present.

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

71. How many nearest neighbors does each silicon atom have in solid Si?

(A) 4 (B) 6 (C) 8 (D) 12

72. Which sample contains the smallest number of atoms?

(A) 1.0 L of Ar at STP
(B) 1.0 L of H_2 at STP
(C) 1.0 L of Ar at 25°C and 760 mm Hg
(D) 1.0 L of H_2 at 0°C and 900 mm Hg

73. Which compound has the lowest normal boiling point?

(A) HF (B) HCl (C) HBr (D) HI

74. Which of the following would lead to an increase in the vapor pressure of a liquid?

I. Increasing the temperature II. Adding a nonvolatile solute
(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

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75. How is the enthalpy of vaporization of a substance related to its enthalpy of fusion?

- (A) The enthalpy of vaporization is greater than the enthalpy of fusion.
- (B) The enthalpy of vaporization is equal to the enthalpy of fusion.
- (C) The enthalpy of vaporization is less than the enthalpy of fusion.
- (D) There is no general relationship between a substance's enthalpy of vaporization and enthalpy of fusion.

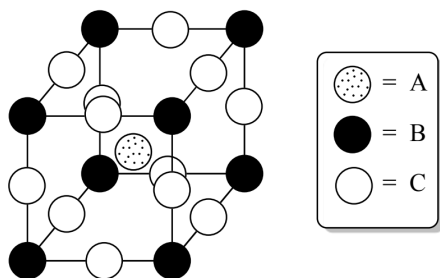
76. A 3.0 L sample of helium gas is stored in a rigid, sealed container at 25 °C and 1.0 atm pressure. The temperature is increased to 125 °C. What is the new pressure of the gas?

- (A) 0.20 atm
- (B) 0.75 atm
- (C) 1.3 atm
- (D) 5.0 atm

77. Diamond is an example of what kind of solid?

- (A) Ionic
- (B) Metallic
- (C) Molecular
- (D) Network covalent

78. The cubic unit cell of a perovskite structure containing atoms of types A, B, and C is illustrated below. What is the empirical formula of this substance?



- (A) ABC (B) ABC₃ (C) AB₄C₆ (D) AB₈C₁₂

79. The kinetic theory of gases assumes all of the following EXCEPT:

- (A) Gases are composed of particles in random, ceaseless motion.
- (B) The sizes of gas particles are negligible compared to the size of the container.
- (C) Gas particles do not attract or repel each other.
- (D) When gas particles collide, kinetic energy is lost.

80. A sample of He gas in a flexible container at room temperature exhibits a certain pressure. What will be the new pressure when the absolute temperature and volume of the container are both halved? The pressure of the He will be

- (A) the same.
- (B) doubled.
- (C) halved.
- (D) quadrupled.

81. The molecules in a sample of pure liquid dichloromethane, CH₂Cl₂, experience which of the following intermolecular forces?

- I dispersion forces
- II dipole-dipole forces
- III hydrogen bonding

- (A) I only
- (B) II only
- (C) I and II only
- (D) I, II, and III

82. Which property best distinguishes metals from other types of solids?

- (A) Metals exhibit three-dimensional order.
- (B) Metals melt at low temperatures.
- (C) Metals have a shiny, silvery-white appearance.
- (D) Metals exhibit three-dimensional electrical conductivity.

83. What type of solid is silicon carbide, SiC?

- (A) ionic
- (B) metallic
- (C) molecular
- (D) network covalent

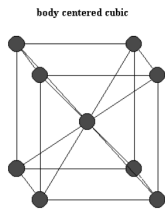
84. The normal boiling point and vapor pressure at 25 °C are measured for liquids in two flasks. Flask A contains pure water and flask B contains a 1.0 M aqueous NaCl solution. Which flask contains the liquid with the higher boiling point? Which flask contains the liquid with the higher vapor pressure?

	<u>Higher boiling point</u>	<u>Higher vapor pressure</u>
(A)	Flask A	Flask A
(B)	Flask A	Flask B
(C)	Flask B	Flask A
(D)	Flask B	Flask B

85. A sample of an ideal gas has a volume of 0.500 L at 25 °C and 1.20 atm pressure. What is its volume at 75 °C and 3.60 atm?

- (A) 0.143 L
- (B) 0.195 L
- (C) 0.500 L
- (D) 1.75 L

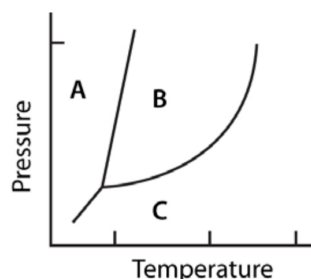
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86. In a mixture of N_2 and O_2 gases, the mol fraction of N_2 is found to be 0.700. The total pressure of the mixture is 1.42 atm. What is the partial pressure of O_2 in the mixture?
- (A) 0.211 atm (B) 0.426 atm
(C) 0.493 atm (D) 0.994 atm
87. The substances below have molar masses that are the same within ± 2 g/mol. Which substance has the lowest boiling point?
- (A) $\text{CH}_3\text{CH}_2\text{CH}_3$ (B) CH_3OCH_3
(C) $\text{CH}_3\text{CH}_2\text{OH}$ (D) CH_3CHO
88. Which statement is correct about the critical point of a phase diagram?
- (A) Solid, liquid and gas are present in equilibrium.
(B) Liquid and vapor are indistinguishable from one another.
(C) Liquid can be produced by a change in pressure.
(D) Vapor can be produced by a change in temperature.
89. When equal volumes of the following pairs of liquids are mixed thoroughly and allowed to stand, which pair is most likely to separate into two layers?
- (A) ethanol and methanol
(B) carbon tetrachloride and methanol
(C) hexane and pentane
(D) carbon tetrachloride and hexane
90. For the same atoms at the lattice points, which lattice exhibits the lowest density?
- (A) body-centered cubic (B) face-centered cubic
(C) hexagonal (D) simple cubic
91. At the molecular level, the factor that determines whether a substance will be a solid, liquid, or gas is the balance between the
- (A) kinetic energies of the molecules and their intermolecular forces.
(B) potential energies of the molecules and their intermolecular forces.
(C) kinetic energies of the molecules and their intramolecular forces.
(D) potential energies of the molecules and their intramolecular forces.
92. A gas sample in a flexible container is maintained at constant pressure while its temperature is increased from 25°C to 75°C . If the initial volume of the gas is 4.2 L, what is the change in volume due to the temperature increase?
- (A) 0.7 L (B) 4.9 L (C) 8.4 L (D) 12.6 L
93. The critical temperature of carbon dioxide is 304.3 K. Which statement is true about the behavior of carbon dioxide above this temperature?
- (A) Solid, liquid and gaseous carbon dioxide are in equilibrium above this temperature.
(B) Liquid and gaseous carbon dioxide are in equilibrium above this temperature.
(C) Liquid carbon dioxide does not exist above this temperature.
(D) Carbon dioxide molecules do not exist above this temperature.
94. Which substance is matched incorrectly with the type of solid it forms?
- (A) ammonium sulfate-ionic solid
(B) lead-metallic solid
(C) potassium chloride-ionic solid
(D) silicon dioxide-molecular solid
95. Metallic sodium has a body-centered cubic unit cell. How many atoms are contained in one unit cell?
- 
- (A) 1 (B) 2 (C) 5 (D) 9
96. Which substance has the lowest boiling point?
- (A) H_2O (B) H_2S (C) H_2Se (D) H_2Te
97. The phase transition from gas to solid is called
- (A) condensation. (B) evaporation.
(C) polymerization. (D) sublimation.
98. A sample of gas occupies a volume of 9.23 L at 345 K and 1.40 atm. What is its volume at 525 K and 3.20 atm?
- (A) 2.65 L (B) 6.14 L (C) 13.9 L (D) 32.1 L

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99. Which substance has the strongest London dispersion forces?
 (A) H_2O (B) H_2S (C) H_2Se (D) H_2Te
100. When the substances Si, KCl, CH_3OH , and C_2H_6 are arranged in order of increasing melting point, what is the correct order?
 (A) Si, KCl, CH_3OH , C_2H_6
 (B) CH_3OH , C_2H_6 , Si, KCl
 (C) KCl, Si, C_2H_6 , CH_3OH
 (D) C_2H_6 , CH_3OH , KCl, Si
101. According to the following information, in what physical state(s) does bromine exist at -7.4°C and 400 mm Hg?
 [Triple point -7.3°C , 44 mm Hg
 Liquid density $3.1\text{ g}\cdot\text{cm}^{-3}$, Solid density $3.4\text{ g}\cdot\text{cm}^{-3}$]
 (A) solid only (B) liquid only
 (C) liquid and solid only (D) gas, liquid, and solid
102. A gas mixture at 27°C and 1 atm contains equal masses of He, H_2 , CO_2 , and CH_4 . How do their molecular velocities compare?
 (A) $\text{He} = \text{H}_2 = \text{CO}_2 = \text{CH}_4$ (B) $\text{He} < \text{H}_2 < \text{CO}_2 < \text{CH}_4$
 (C) $\text{H}_2 < \text{He} < \text{CH}_4 < \text{CO}_2$ (D) $\text{CO}_2 < \text{CH}_4 < \text{He} < \text{H}_2$
103. A 2.00 L balloon at 20.0°C and 745 mmHg floats to an altitude where the temperature is 10.0°C and the air pressure is 700 mmHg. What is the new volume of the balloon?
 (A) 0.94 L (B) 1.06 L (C) 2.06 L (D) 2.20 L
104. Which family of elements has solid, liquid and gaseous members at 25°C and 1 atm pressure?
 (A) alkali metals (Li – Cs) (B) pnictogens (N – Bi)
 (C) chalcogens (O – Te) (D) halogens (F – I)
105. A gas diffuses one-third as fast as O_2 at 100°C . This gas could be
 (A) He ($M = 4$) (B) $\text{C}_2\text{H}_5\text{F}$ ($M = 48$)
 (C) C_7H_{12} ($M = 96$) (D) C_5F_{12} ($M = 288$)
106. Moist air is less dense than dry air at the same temperature and barometric pressure. Which is the best explanation for this observation?
 (A) H_2O is a polar molecule but N_2 and O_2 are not.
 (B) H_2O has a higher boiling point than N_2 or O_2 .
 (C) H_2O has a lower molar mass than N_2 or O_2 .
 (D) H_2O has a higher heat capacity than N_2 or O_2 .

107. Under certain conditions CO_2 melts rather than sublimes. To which transition in the phase diagram does this change correspond?



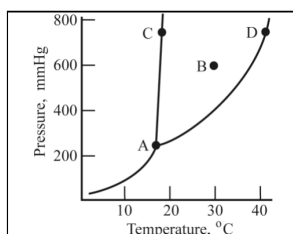
- (A) $A \rightarrow B$ (B) $A \rightarrow C$ (C) $B \rightarrow C$ (D) $C \rightarrow B$
108. The critical temperature of water is the
 (A) temperature at which solid, liquid and gaseous water coexist.
 (B) temperature at which water vapor condenses.
 (C) maximum temperature at which liquid water can exist.
 (D) minimum temperature at which water vapor can exist.
109. A sample of gas at 273 K has a pressure of P_1 and a volume of V_1 . When the pressure is changed to P_2 , what is the volume V_2 ? (Assume the temperature remains constant.)
 (A) $\frac{P_1 P_2}{V_1}$ (B) $\frac{P_1 V_1}{P_2}$
 (C) $\frac{P_2 V_1}{P_1}$ (D) $\frac{P_2}{P_1 V_1}$
110. How do the number of molecules, n , in 1.0 L of each of the following gases; CH_4 , N_2 , CO_2 , compare at 1 atm and 25°C ?
 (A) $n_{\text{CH}_4} < n_{\text{CO}_2} < n_{\text{N}_2}$ (B) $n_{\text{N}_2} < n_{\text{CO}_2} < n_{\text{CH}_4}$
 (C) $n_{\text{CO}_2} < n_{\text{CH}_4} < n_{\text{N}_2}$ (D) $n_{\text{CH}_4} = n_{\text{CO}_2} = n_{\text{N}_2}$
111. Solid sodium acetate, $\text{NaC}_2\text{H}_3\text{O}_2$, is what type of solid?
 (A) ionic (B) metallic
 (C) molecular (D) network covalent

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112. Which substance has the highest vapor pressure at 25°C?

- (A) methanol, CH_3OH
- (B) ethanol, $\text{CH}_3\text{CH}_2\text{OH}$
- (C) 1-propanol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- (D) 1-butanol, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$

113. Which point on the phase diagram represents the normal boiling point?



- (A) point A
- (B) point B
- (C) point C
- (D) point D

114. What types of intermolecular forces are exerted by CH_3Cl molecules in the liquid phase?

- I. dipole-dipole forces
- II. hydrogen bonding
- III. London dispersion forces

- (A) I only
- (B) II only
- (C) I and III only
- (D) II and III only

115. Liquids exhibit all these properties **except**

- (A) definite volume.
- (B) definite shape.
- (C) incompressibility.
- (D) slow diffusion.

116. Which statement is true about a substance that is subjected to a lower external pressure at a constant temperature?

- (A) A liquid will boil at a lower temperature.
- (B) A liquid will exhibit a lower vapor pressure.
- (C) A gas in an insulated container will change into a liquid.
- (D) A gas in a nonrigid container will exhibit a smaller volume

117. A flask contains a mixture of Ne(g) and Ar(g) . There are 0.250 mol of Ne(g) which exerts a pressure of 205 mmHg. If the Ar(g) exerts a pressure of 492 mmHg, what mass of Ar(g) is in the flask?

- (A) 4.16 g
- (B) 12.1 g
- (C) 24.0 g
- (D) 95.9 g

118. A gas in a closed, flexible container is slowly cooled from 50 °C to 25 °C. What is the ratio of the final volume of the gas to its initial volume? Assume ideal behavior.

- (A) 2/1
- (B) 1.08/1
- (C) 0.923/1
- (D) 0.5/1

119. The value of which property decreases with an increase in the strength of intermolecular forces?

- (A) viscosity
- (B) boiling point
- (C) surface tension
- (D) vapor pressure

120. A white solid dissolves in water to form a solution that does not conduct electricity. What type of bonding is most likely present in the solid?

- (A) ionic
- (B) metallic
- (C) network covalent
- (D) polar covalent

121. All of these are characteristics of MOST ionic compounds in the solid phase EXCEPT

- (A) high electrical conductivity
- (B) high melting point
- (C) solubility in water
- (D) insolubility in organic solvents

122. Which noble gas effuses approximately twice as fast as Kr?

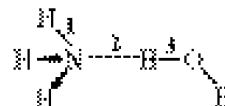
Molar Mass (g/mol)	
Ne	20.18
Ar	39.95
Kr	83.80
Xe	131.3
Rn	222

- (A) Ne
- (B) Ar
- (C) Xe
- (D) Rn

123. Rank the enthalpies of fusion, sublimation and vaporization for water.

- (A) sublimation = vaporization = fusion
- (B) vaporization < sublimation < fusion
- (C) fusion < sublimation < vaporization
- (D) fusion < vaporization < sublimation

124. In this diagram, which bonds represent hydrogen bonds?



- (A) 1 only
- (B) 2 only
- (C) 1 and 3 only
- (D) 1, 2 and 3

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125. A sample of C_2H_6 gas initially at $50^\circ C$ and 720 mmHg is heated to $100^\circ C$ in a container of constant volume. What is the new pressure (in mmHg)?

(A) 360 (B) 623 (C) 831 (D) 1440

126. All of the following properties of liquids increase with increasing strengths of intermolecular forces EXCEPT

(A) boiling point (B) enthalpy of vaporization
(C) vapor pressure (D) viscosity

127. Which pure compounds form intermolecular hydrogen bonds?

I. HF
II. H_2S
III. CH_4

(A) I only (B) II only
(C) I and II only (D) II and III only

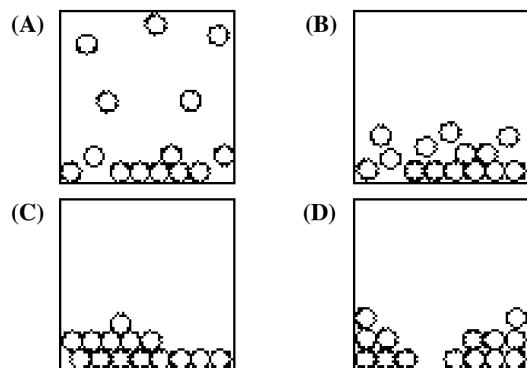
128. What is the molar mass of a gas if 10.0 grams of it occupy 4.48 liters at 273 K and 101.3 kPa (1.00 atm)?

(A) 2.00 g/mol (B) 25.0 g/mol
(C) 50.0 g/mol (D) 100. g/mol

129. Which process requires the greatest amount of energy for 1 mole of H_2O ?

(A) breaking the O-H bonds (B) melting
(C) evaporating (D) subliming

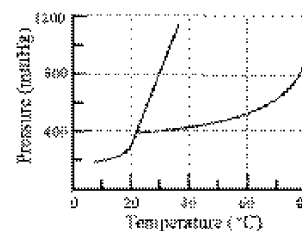
130. If the circles represent molecules, which diagram provides the best molecular level representation of a pure solid in the process of melting?



131. Which substance boils at the highest temperature?

(A) CH_3Cl (B) CH_2Cl_2
(C) $CHCl_3$ (D) CCl_4

132. What can be concluded about the substance represented by this phase diagram?



(A) The normal boiling point of the liquid is $80^\circ C$.
(B) The solid is more dense than the liquid.
(C) The solid sublimates at temperatures above $20^\circ C$.
(D) The vapor can be converted to liquid by compressing it at temperatures below $20^\circ C$.

133. A gas mixture at $27^\circ C$ and 760 mm Hg contains 1.0 g each of He, H_2 , N_2 and CO_2 . How do their average molecular speeds compare?

(A) $He = H_2 = N_2 = CO_2$ (B) $CO_2 < H_2 = N_2 < He$
(C) $He < H_2 < N_2 < CO_2$ (D) $CO_2 < N_2 < He < H_2$

134. A 2.00 liter evacuated container has a mass of 1050.0 g. When the container is filled with an unknown gas at 800. mm Hg pressure and $25.0^\circ C$ the mass is 1052.4 g. What is the molar mass of the gas (in $g \cdot mol^{-1}$)?

(A) 28 (B) 31 (C) 54 (D) 56

135. A mixture of 0.100 mol of N_2 and 0.200 mol of O_2 is

Vapor pressure @ $22^\circ C$	
H_2O	22 mmHg

collected over H_2O at an atmospheric pressure of 750. mm Hg and a temperature of $22^\circ C$. What is the partial pressure (in mmHg) of O_2 in this mixture?

(A) 478 (B) 485 (C) 500 (D) 515

136. A solid is insoluble in water, does not conduct electricity, and does not melt below $1000^\circ C$. This solid could be

(A) Pt. (B) SiC. (C) CsCl. (D) $C_{10}H_{22}$.

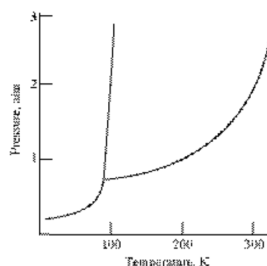
137. Which property(ies) of a liquid increases when the temperature is raised?

I. vapor pressure
II. surface tension

(A) I only (B) II only
(C) Both I and II (D) Neither I nor II

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138. Which statement is correct about the substance represented by this phase diagram?



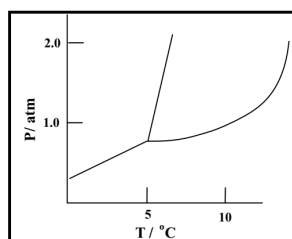
- (A) The solid sublimates at 1 atm pressure.
 (B) Its normal boiling point is above 300K.
 (C) It exists as a liquid at 25 °C and 1 atm pressure.
 (D) The density of the solid is greater than that of the liquid.

139. A sample of oxygen gas and a sample of an unknown gas are weighed separately in the same evacuated flask. Use the data given to find the molar mass of the unknown gas (assume experiments are carried out at the same pressure and temperature).

Mass of evacuated flask	124.46 g
Mass of flask + oxygen	125.10 g
Mass of flask + unknown gas	125.34 g

- (A) 22 g/mol (B) 38 g/mol
 (C) 44 g/mol (D) 84 g/mol
140. Which pair of gases has the same average rate of diffusion at 25 °C?
- (A) He and Ne (B) N₂ and O₂
 (C) N₂O and CO₂ (D) NH₃ and HCl

141. According to the phase diagram shown, in what state does the represented substance exist at 1.0 atm and 0.0 °C?



- (A) solid only (B) liquid only
 (C) gas only (D) solid and liquid only

142. What is the most effective way to condense a gas?

- (A) Decrease the temperature and increase the pressure.
 (B) Decrease the temperature and decrease the pressure.
 (C) Increase the temperature and decrease the pressure.
 (D) Increase the temperature and increase the pressure.

143. Which liquid has the highest vapor pressure at 25 °C?

- (A) butane, C₄H₁₀ (B) glycerol, C₃H₅(OH)₃
 (C) octane, C₈H₁₈ (D) propanol, C₃H₇OH

144. Which oxide has the highest melting point?

- (A) H₂O (B) NO₂ (C) SO₂ (D) SiO₂

145. A gas has a volume of 6.0 L at a pressure of 0.80 atm. What is the volume if the pressure is changed to 0.20 atm at constant temperature?

- (A) 1.5 L (B) 3.0 L (C) 12 L (D) 24 L

146. Which will increase the vapor pressure of a liquid?

- 1 increase in temperature
 2 increase in surface area

- (A) 1 only (B) 2 only
 (C) Both 1 and 2 (D) Neither 1 nor 2

147. What pressure (in atm) will be exerted by a 1.00 g sample of methane, CH₄, in a 4.25 L flask at 115 °C?

- (A) 0.139 (B) 0.330 (C) 0.467 (D) 7.50

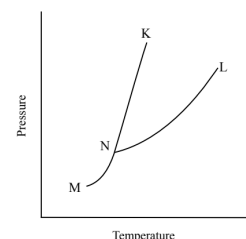
148. The lowest melting points overall occur for members of which class of solids?

- (A) ionic (B) metallic
 (C) molecular (D) network covalent

149. What are the strongest intermolecular force between neighboring carbon tetrachloride, CCl₄, molecules?

- (A) dipole-dipole forces (B) dispersion forces
 (C) hydrogen bonds (D) covalent bonds

150. According to the phase diagram shown, where does a mixture of solid and liquid exist at equilibrium?



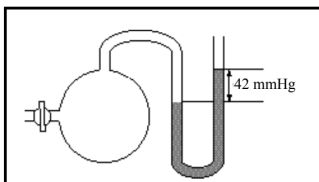
- (A) along line MN (B) along line KN
 (C) along line LN (D) in the region KNL

151. A sample of neon gas has a volume of 248 mL at 30.°C and a certain pressure. What volume would it occupy if it were heated to 60.°C at the same pressure?

- (A) 226 mL (B) 273 mL
 (C) 278 mL (D) 496 mL

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152. A gas is collected in the flask shown here. What is the pressure exerted by the gas if the atmospheric pressure is 735 mmHg?



- (A) 42 mmHg (B) 693 mmHg
(C) 735 mmHg (D) 777 mmHg
153. Helium is often found with methane, CH_4 . How do the diffusion rates of helium and methane compare at the same temperature? Helium diffuses
- (A) sixteen times as fast as methane.
(B) four times as fast as methane.
(C) twice as fast as methane.
(D) at the same rate as methane.
154. Which substance contains individual molecules in the solid?
- (A) graphite (B) iodine
(C) mercury (D) silicon carbide
155. The compounds C_3H_8 , $\text{CH}_3\text{CH}_2\text{OH}$, and CH_3OCH_3 have very similar molar masses. When they are arranged in order of *increasing* strength of their intermolecular forces, what is the correct order?
- (A) C_3H_8 , CH_3OCH_3 , $\text{CH}_3\text{CH}_2\text{OH}$
(B) $\text{CH}_3\text{CH}_2\text{OH}$, CH_3OCH_3 , C_3H_8
(C) CH_3OCH_3 , C_3H_8 , $\text{CH}_3\text{CH}_2\text{OH}$
(D) $\text{CH}_3\text{CH}_2\text{OH}$, C_3H_8 , CH_3OCH_3
156. Which property does *not* indicate strong intermolecular forces?
- (A) high enthalpy of vaporization
(B) high viscosity
(C) high critical temperature
(D) high vapor pressure
157. Which terms are matched correctly?
- (A) high vapor pressures high ΔH_{vap} values
(B) high ΔH_{vap} values low boiling points
(C) low vapor pressure high boiling points
(D) low boiling points low vaporization rates

158. Three balloons are filled with the same number of atoms of He, Ar, and Xe, respectively. Which statement is true under the same conditions of temperature and pressure?

- (A) The balloons contain the same mass of gas.
(B) All balloons have the same volume.
(C) The densities of the three gases are the same.
(D) The average speed of the different types of atoms is the same.

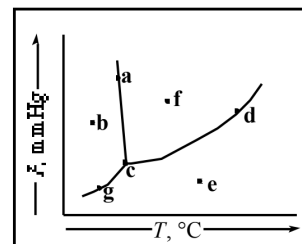
159. A 0.239 g sample of a gas in a 100-mL flask exerts a pressure of 600 mmHg at 14 °C. What is the gas?

- (A) chlorine (B) nitrogen
(C) krypton (D) xenon

160. Which parameter affects the vapor pressure of a liquid?

- (A) volume of the liquid
(B) surface area of the liquid
(C) volume of space above the liquid
(D) temperature of the liquid

161. Which points in this phase diagram represent conditions of temperature and pressure where liquid will be present?



- (A) a, b, and g only (B) a, c, d and g only
(C) a, c, d and f only (D) c, d, e and g only

162. Which compound is expected to be the most soluble in water at 25 °C?

- (A) $\text{N}_2(\text{g})$ (B) $\text{O}_2(\text{g})$
(C) $(\text{C}_2\text{H}_5)_2\text{NH}(\text{l})$ (D) $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5(\text{l})$

163. A hard, crystalline solid with a high melting point does not conduct electricity in any phase. This solid is most likely

- (A) an ionic solid.
(B) a metallic solid.
(C) a molecular solid.
(D) a network covalent solid.

164. Under which conditions will a gas behave most ideally?

- (A) low P and high T (B) low P and low T
(C) high P and low T (D) high P and high T

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165. The mass of 560 cm^3 of a gas at 0°C and 1 atm is 1.60 g. Which gas could it be?
- (A) O_2 (B) CO_2 (C) SO_2 (D) Cl_2
166. Oxygen, which is 16 times as dense as hydrogen, diffuses
- (A) 1/16 times as fast. (B) 1/4 times as fast.
(C) 4 times as fast. (D) 16 times as fast.
167. How is the vapor pressure of a liquid in a closed container affected when the quantity of liquid is doubled at constant temperature?
- (A) The vapor pressure increases.
(B) The vapor pressure decreases.
(C) The vapor pressure stays the same.
(D) The vapor pressure may increase or decrease, depending on the liquid.
168. A low molar heat of fusion is expected for a solid that is
- (A) ionic. (B) metallic.
(C) molecular. (D) network covalent.
169. Which characteristic is most useful for determining that a substance is a metal?
- (A) conductivity (B) hardness
(C) melting point (D) X-ray pattern